

$$\therefore v = 3.45 \times 10^6 \text{ m s}^{-1}$$

- 6 (a) Photoelectric effect is the emission of electrons from a metal surface when it is exposed to electromagnetic radiation of sufficiently high frequency.
- (b) Photons interact with the electrons in the metal, being either completely absorbed or not at all. In order for photoelectric emission to take place, the individual photon's energy must be greater than a minimum amount, known as the metal's work function energy, $\phi = hf_0$, where f_0 is the threshold frequency. If the radiation has frequency $f < f_0$, then the photon's energy $E < \phi$, where $E = hf$ is the energy of a photon. Even if the intensity of the radiation increases, the number of photons arriving per unit time merely increases. However, the energy E carried by a photon is still the same. Since the energy of different photons cannot be cumulated, no photoelectric emission takes place.